Version 2 evaluation report

Data: `data/inventory_med_imerg.csv.gz` (overlapping areas merged): 216171 rows, 13 storm events, 213489 segments, 23340 documented flooded rows (10.80%).

Protocol: 13-fold cross-validation grouped by storm event (a storm is never in train and test). Inside each training fold, calibration and level thresholds use out-of-fold predictions, also grouped by event. Levels follow an alert budget inside each mapped area of the test event (schema.ALERT_SHARE: High = riskiest 2 % of each area), so every model raises the same number of alerts and recall and false-alarm rate compare directly. Rainfall at ETA is replaced by a forecast whose error grows with lead time.

Results against documented floods (lead 20 min)

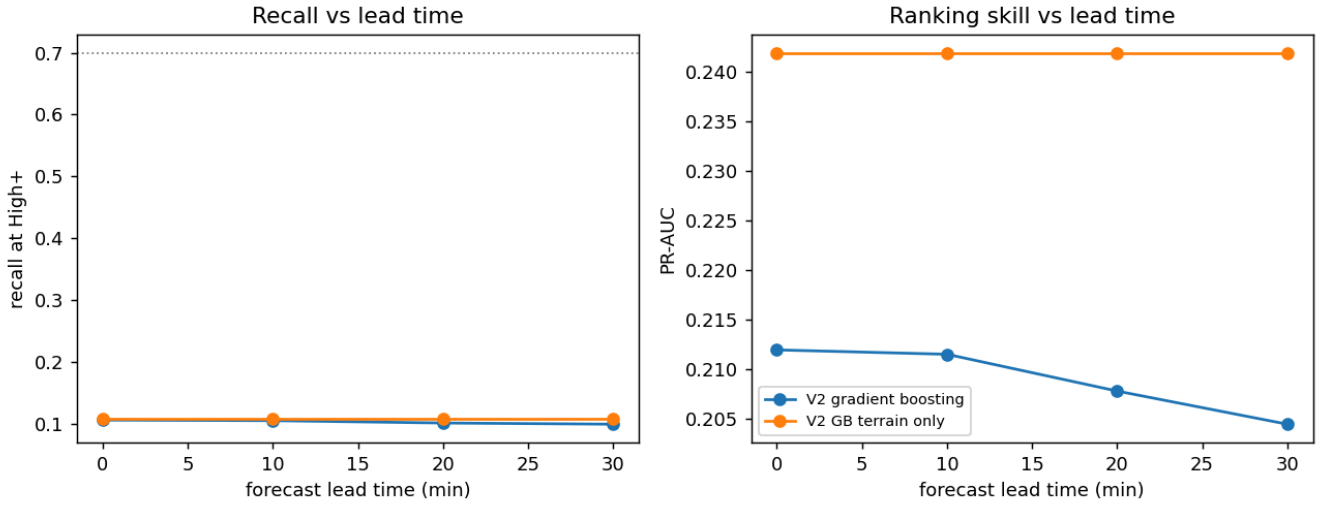
Model	Recall High+	False-alarm rate	Alerts /1000	ROC-AUC	PR-AUC	Recall top 2 %	Brier	ECE
V2 gradient boosting	0.101 ± 0.116	0.736 ± 0.223	19.8	0.739 ± 0.175	0.208 ± 0.117	0.084 ± 0.096	0.1016	0.1101
V2 GB terrain only	0.108 ± 0.125	0.718 ± 0.226	19.8	0.738 ± 0.186	0.242 ± 0.153	0.100 ± 0.118	0.1020	0.1113

Event-level results (lead 20 min)

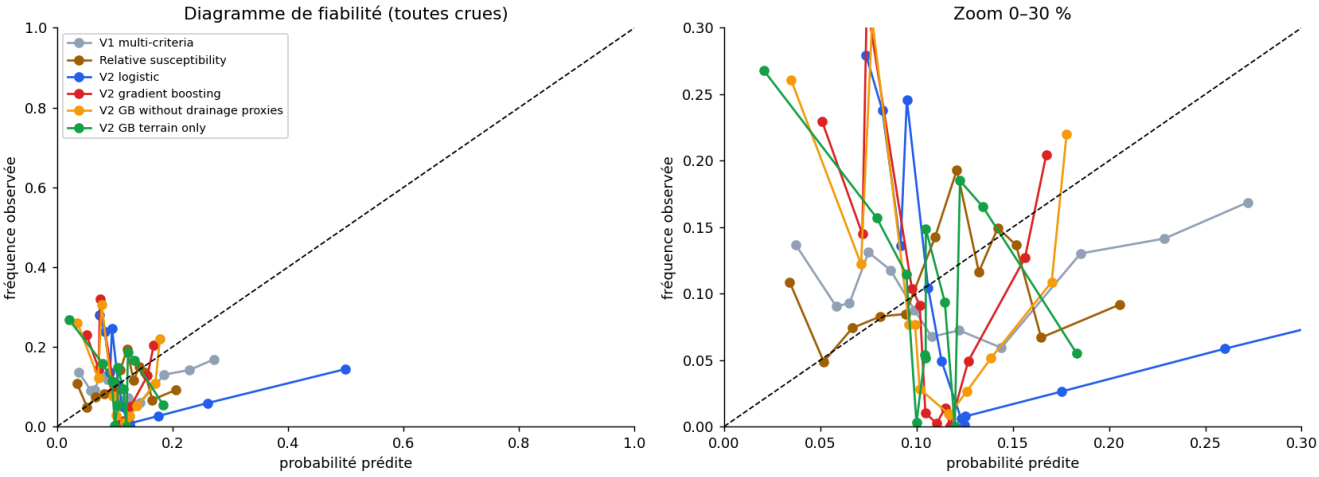
One row per segment and storm: alert if the maximum level during the storm is High or above, flooded if the segment flooded at any time. This matches how press and civil-protection reports are dated.

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Lead time



Calibration

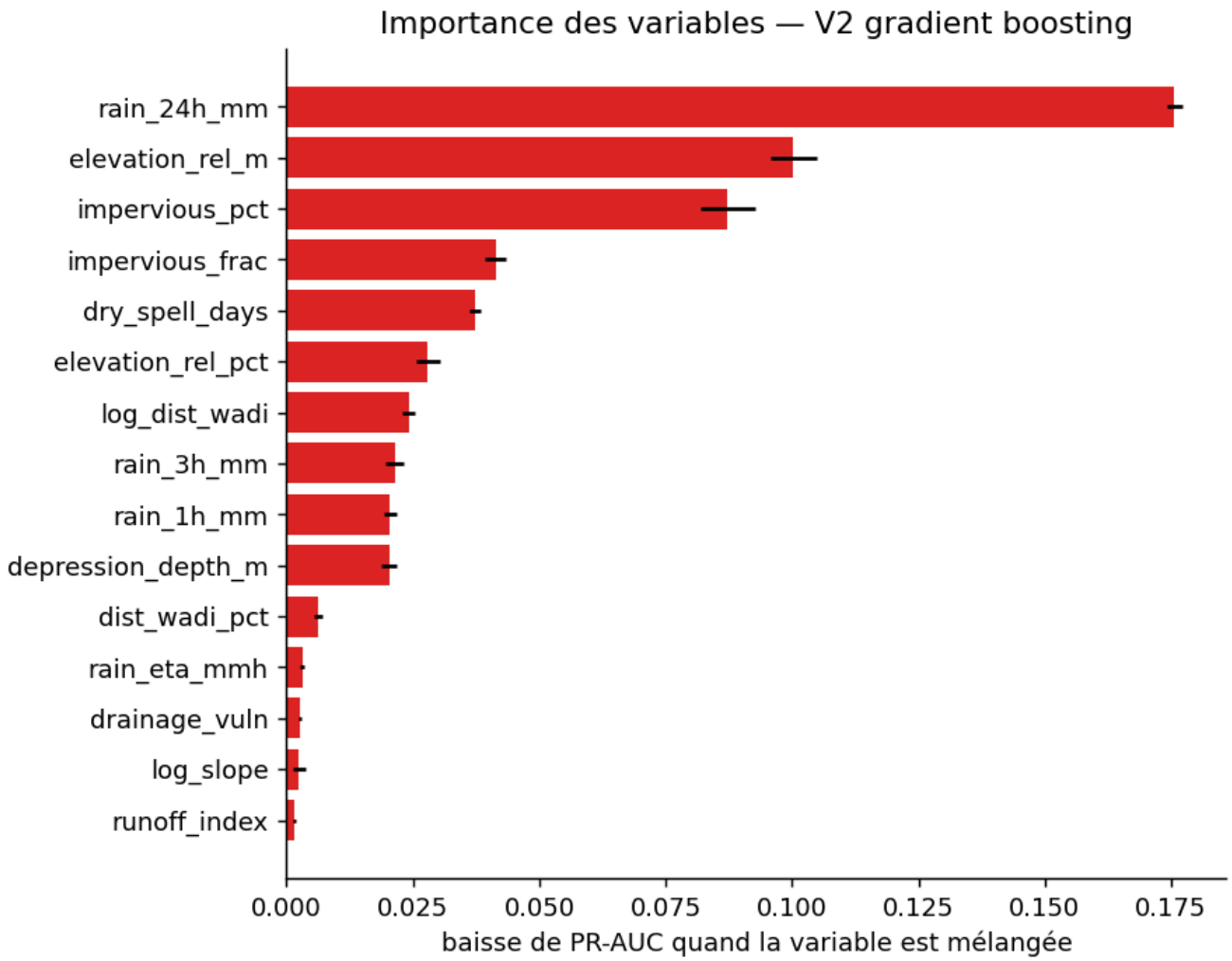


Drainage factor usefulness (section 8)

Among documented flooded rows, rainfall intensity at flooding for the top third of estimated drainage vulnerability: median 15.9 mm/h (n=7785); bottom third: 18.1 mm/h (n=7783). One-sided Mann-Whitney p = 4.1e-53.

This test alone does not prove the drainage hypothesis: high-vulnerability segments are also the most impervious, and impervious roads flood at lower rainfall whatever the state of their drains. The ablation row *V2 GB without drainage proxies* is the stronger test: the proxies are useful only if removing them lowers PR-AUC or raises the false-alarm rate on held-out storms (that row was not run here: the ablation is in `pipeline_v2.ipynb`, section 9.5).

Feature importance (final gradient-boosting model)



Feature	Drop in PR-AUC
rain_24h_mm	0.1795
elevation_rel_m	0.1006
impervious_pct	0.0851
impervious_frac	0.0445
dry_spell_days	0.0324
elevation_rel_pct	0.0285
log_dist_wadi	0.0266
rain_1h_mm	0.0208
depression_depth_m	0.0203
rain_3h_mm	0.0196

Deployed thresholds (calibrated probability)

Model	Moderate	High	Very high
V2 gradient boosting	0.123	0.227	0.345
V2 logistic	0.107	0.117	0.143